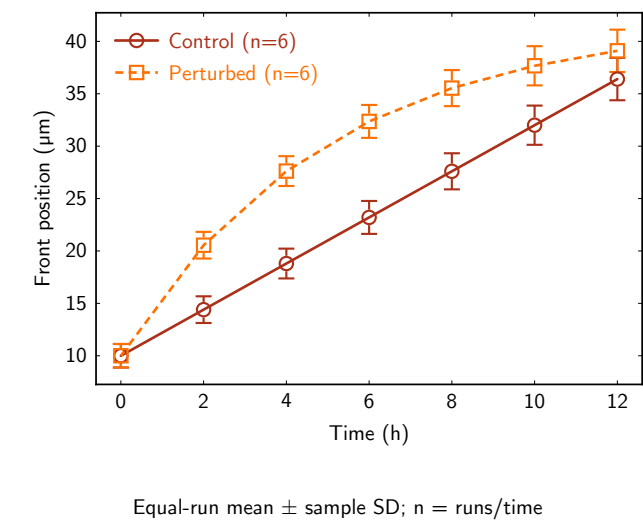


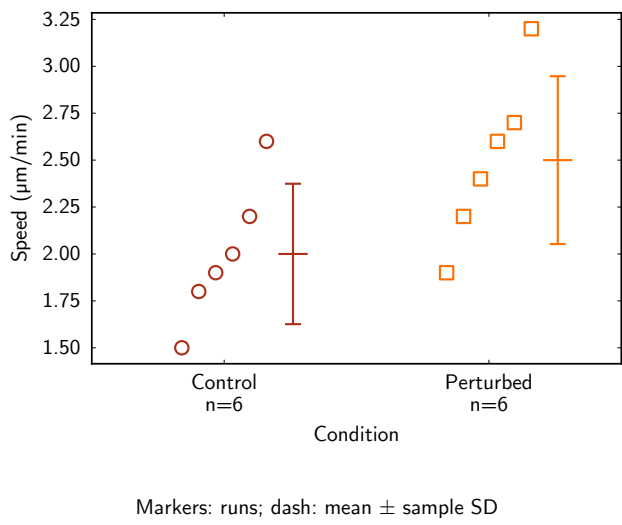
Quantitative plots, with restraint.

Eight original examples · consistent font sizes · PDF review gallery

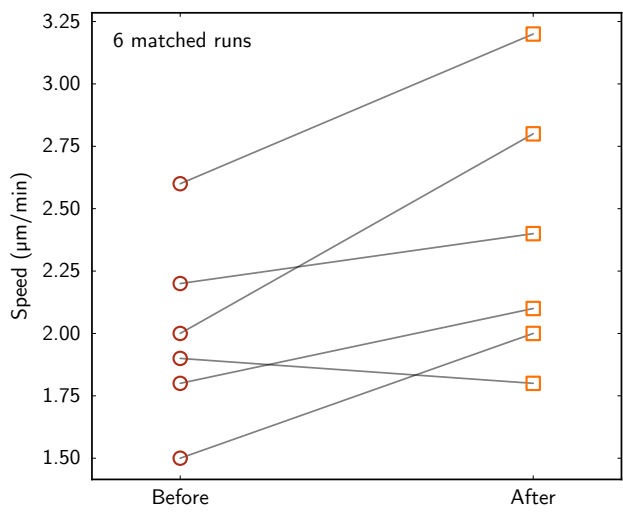
01 Time courses



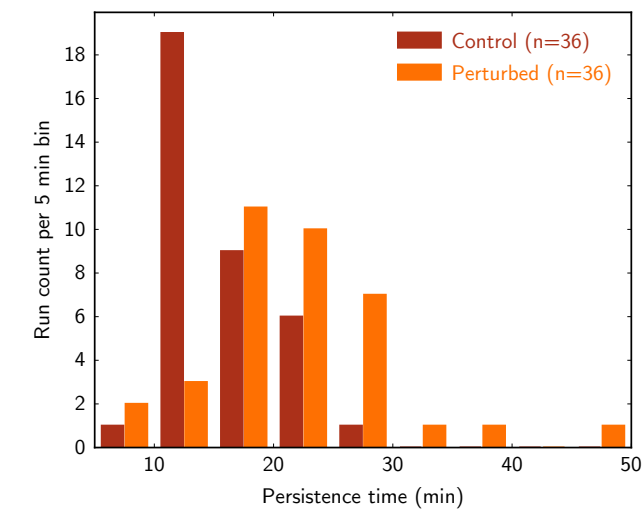
02 Replicate comparisons



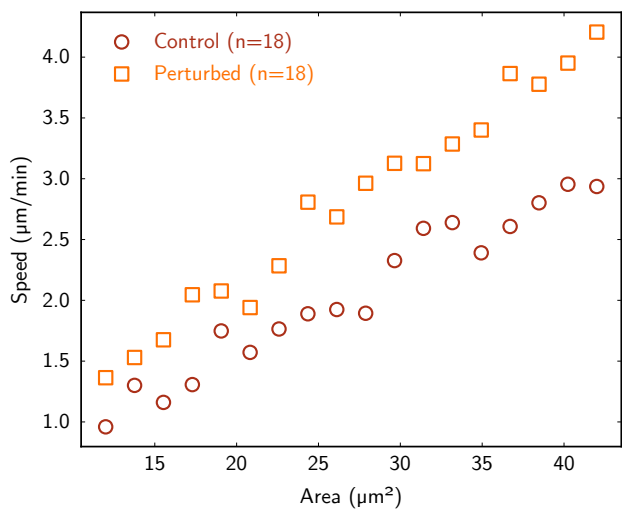
03 Paired changes



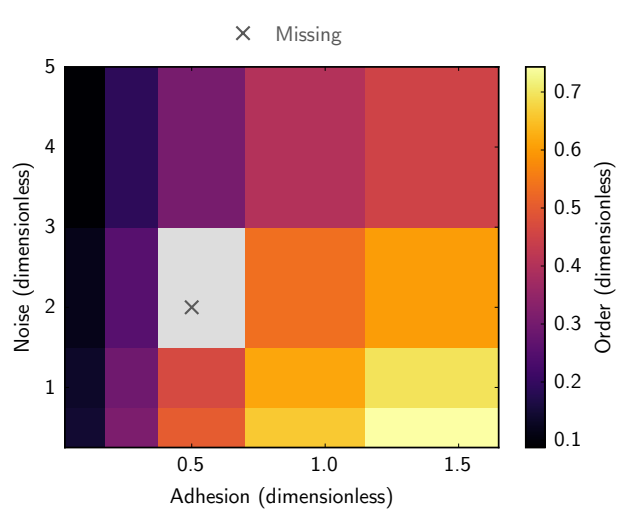
04 Distributions



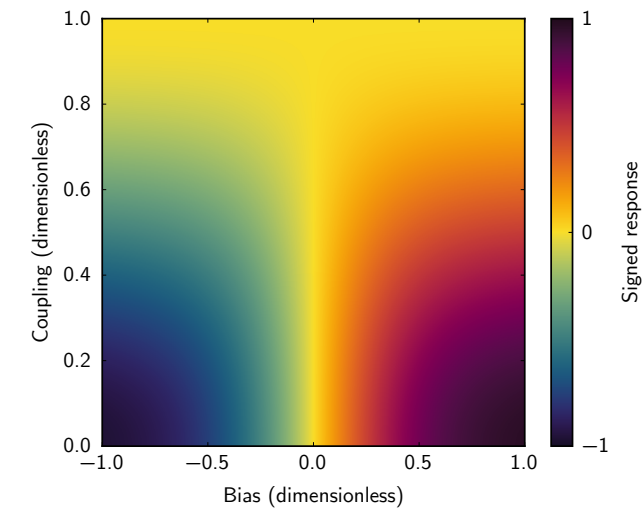
05 Relationships



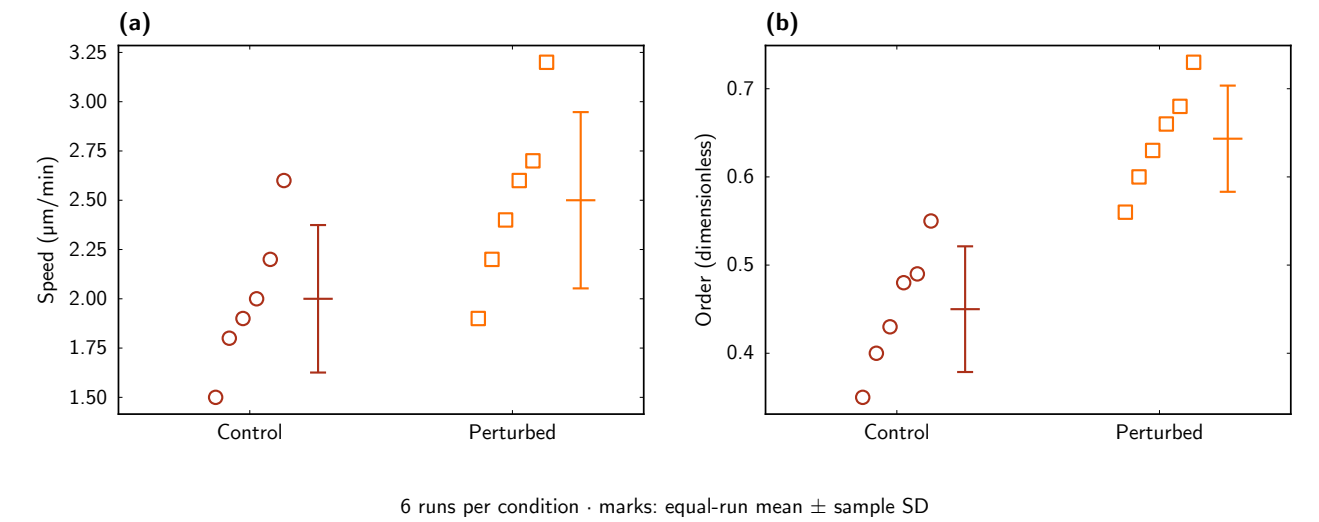
06 Parameter maps



07 Continuous functions

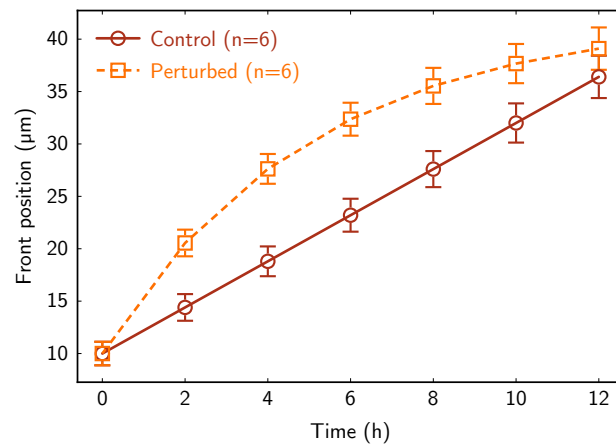


08 Related metrics



Time courses

Follow a quantity through time.



Equal-run mean $\pm$ sample SD; n = runs/time

READ THE DATA

Each hollow marker is the equal-run mean of six saved run values. Capped error bars show sample SD, not a confidence interval.

DESIGN DECISION

All point markers use the same size and light outline. Error bars and legend text match their condition's color; there is no shaded band.

KEEP THE SCIENCE

Do not bridge an unmeasured time point or count individual cells as independent runs.

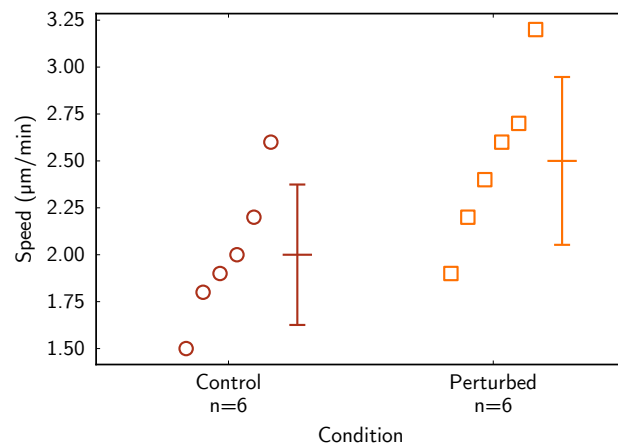
SOURCE

assets/examples/timecourse/source.py

Teaching table: assets/examples/timecourse/data.csv

Replicate comparisons

Show variation before summarizing it.



Markers: runs; dash: mean $\pm$ sample SD

READ THE DATA

Each hollow point is one run. Beside the points, a horizontal mark is the equal-run mean and whiskers show sample SD. n counts measured runs.

DESIGN DECISION

Modest horizontal offsets separate observations without altering their values. Shape reinforces condition identity.

KEEP THE SCIENCE

Use a point comparison for a focused scale; do not imply differences through truncated bar lengths.

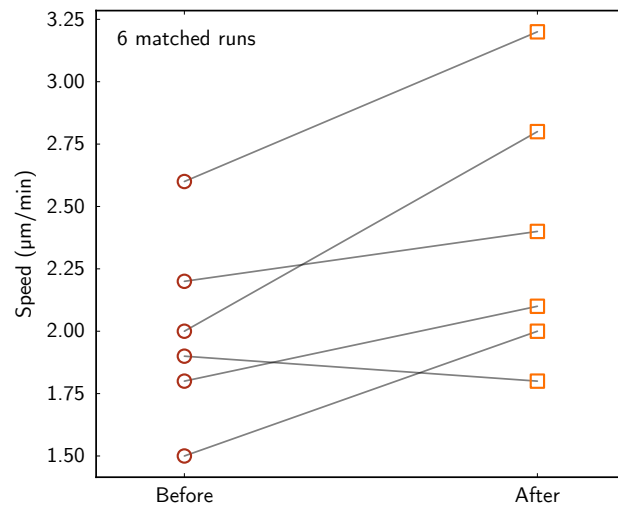
SOURCE

assets/examples/replicate-comparison/source.py

Teaching table: assets/examples/replicate-comparison/data.csv

Paired changes

Keep the same run connected.



READ THE DATA

Each line connects two values belonging to one run. All six pairs are shown, including the run whose value falls.

DESIGN DECISION

Neutral connectors carry pairing. Two restrained endpoint colors show measurement state; no fitted trend is added.

KEEP THE SCIENCE

Pair by the run key. Never connect independently sorted arrays or silently drop unmatched runs.

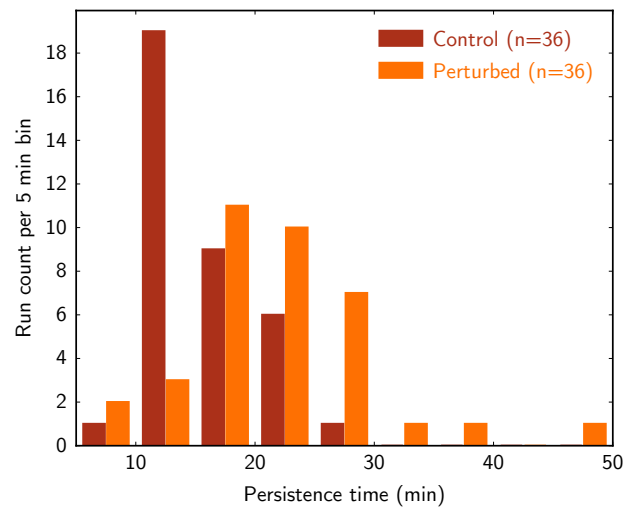
SOURCE

`assets/examples/paired-change/source.py`

Teaching table: `assets/examples/paired-change/data.csv`

Distributions

Count observations in clear, shared bins.



READ THE DATA

Bar heights count runs in common five-minute bins; $n=36$ per condition. Control is the left bar and Perturbed the right bar in each pair.

DESIGN DECISION

Separate bars use solid group-color fills with clear boundaries and small gaps. Legend text matches each bar's color.

KEEP THE SCIENCE

Use the same bin edges and zero baseline for both groups. Keep every value, including tails; do not confuse counts with probability density.

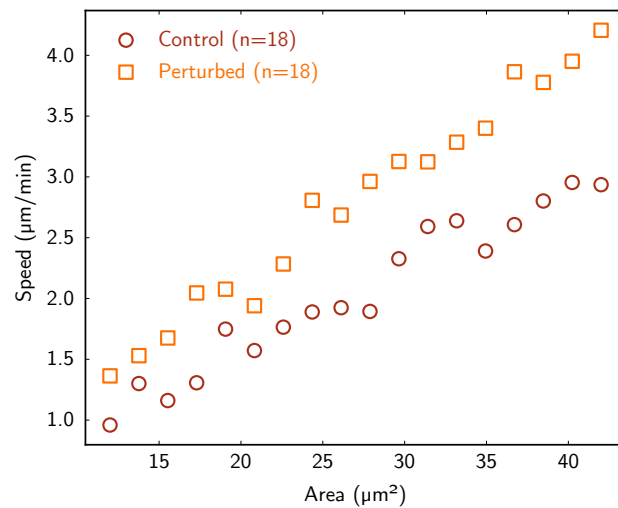
SOURCE

`assets/examples/distribution/source.py`

Teaching table: `assets/examples/distribution/data.csv`

Relationships

Show paired observables directly.



READ THE DATA

Each point contains area and speed from the same run; there are 18 runs per condition. No fitted line or inferential statistic is implied.

DESIGN DECISION

Hollow markers share the same size and light outline as the other plots. Shape and matching legend text identify condition; units name both coordinates.

KEEP THE SCIENCE

Align both measurements by scientific identity. Equal array lengths do not establish pairing.

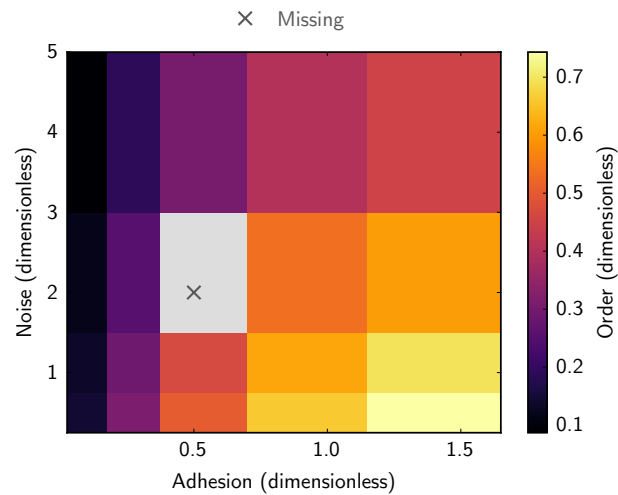
SOURCE

assets/examples/scatter/source.py

Teaching table: assets/examples/scatter/data.csv

Parameter maps

Keep coordinates and missingness visible.



READ THE DATA

Each measured parameter pair supplies one descriptive order value. Unequal coordinate spacing is retained. Gray and a cross mark one missing value.

DESIGN DECISION

The package's sequential colormap encodes magnitude. Display-cell boundaries are midpoints between sampled coordinates.

KEEP THE SCIENCE

Cells summarize sampled values, not a continuous interpolated phase diagram. Missing is not zero.

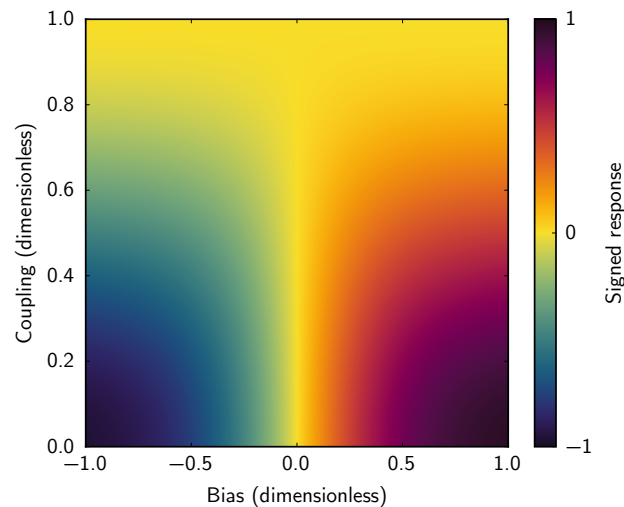
SOURCE

`assets/examples/parameter-map/source.py`

Teaching table: `assets/examples/parameter-map/data.csv`

Continuous functions

Render a continuous field smoothly.



READ THE DATA

An original analytic function is evaluated on a dense 201 by 101 grid. Its signed color scale is centered on zero and fixed to $[-1, 1]$.

DESIGN DECISION

Bilinear image rendering follows the minimalist example. The field stays continuous without cell outlines or a grid of numeric labels.

KEEP THE SCIENCE

Smooth display is appropriate for this known continuous function. Do not smooth sparse parameter samples or interpolate across missing observations.

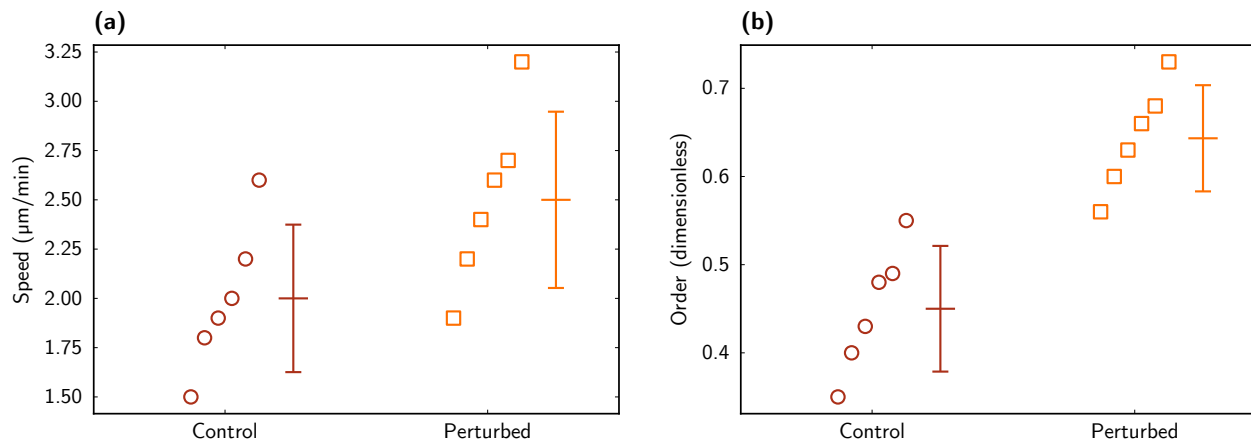
SOURCE

assets/examples/continuous-field/source.py

Teaching table: assets/examples/continuous-field/data.csv

Related metrics

Keep units separate and conditions consistent.



6 runs per condition · marks: equal-run mean $\pm$ sample SD

READ THE DATA

Two metrics use the same six runs per condition. Hollow points are run values; matching-color marks and whiskers beside them show mean and sample SD.

DESIGN DECISION

Aligned panels repeat the same condition colors and markers. Each metric retains its own labeled scale and units.

KEEP THE SCIENCE

Do not use a dual axis to imply a relationship or merge quantities with different units into one scale.

SOURCE

assets/examples/metric-comparison/source.py

Teaching table: assets/examples/metric-comparison/data.csv

SYNTHETIC DATA · Design specimen, not a scientific result